Fiscal Year:	FY 2020	**Task Last Updated:**	FY 07/22/2019
PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2021 per PI (Ed., 7/8/19) NOTE: Extended to 9/30/2019 per PI (Ed., 7/5/16) NOTE: Extended to 10/1/16 (Ed., 7/31/13) NOTE: To be extended past 10/1/2013 per PI (Ed., 7/24/13)		
Key Personnel Changes/Previous PI:	July 2019 report: Prof. William L. Johnson has moved from Caltech Professor to Emeritus Caltech Professor.		
COI Name (Institution):	Johnson, William Ph.D. (California Institute of Technology) Roberts, Scott Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Amorphous metals (AMs), also known as Bulk Metallic Glasses (BMGs), are a unique class of metal alloys known for their non-crystalline microstructures, unique mechanical properties, and low melting temperatures. BMG matrix composites are amorphous metals reinforced with crystalline phases to improve mechanical properties such as toughness, strength and wear resistance. When BMGs become crystalline, their toughness and wear resistance is normally severely decreased. By strategically adding crystalline particles, such as dense tungsten (W), within the crystalline BMG matrix, the wear resistance can be remarkably improved. However, sedimentation dramatically affects how these composites can be processed. The microgravity environment aboard the International Space Station (ISS) will be utilized to perform a semi-solid, isothermal (or mushy) hold on the composites between the solidus and the liquidus to selectively precipitate out dendrites into the BMG matrix, resulting in alloys that have wear resistance and toughness that cannot be matched through terrestrial processing. The investigators propose to study the nucleation, growth, directionality, and cooling rate dependence of crystalline dendrites evolving in a glass-forming eutectic matrix. The ground-based research effort will be carried out utilizing laboratories at the Jet Propulsion Laboratory and the California Institute of Technology, including the High Vacuum ElectroStatic Levitator platform (HVESL) at Caltech. The investigators propose a four-component ground-based project consisting of (1) directional solidification experiments utilizing Moving Mushy Zone (MMZ) and Bridgman solidification, (2) morphological characterization of dendrite growth and orientation during "mushy" (or "semi-solid") processing, (3) mechanical testing and development of structure-property relations, and (4) dendrite formation and growth during containerless processing utilizing the HVESL. After completion of a ground-based research program, the investigators will extend the experimentation to flight-based microgravity research aboard the International Space Station (ISS), the aim of which is to study gravity-driven convection arising from density gradients in the microstructure. The flight experiment objectives are to (1) use the Low Gradient Furnace (LGF) to complete Bridgman crystal growth on composites in an attempt to grow dendrites along the translational axis, (2) use the Solidification and Quenching Furnace to coarsen and homogenize dendrite microstructures using mushy-zone processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
Task Progress:	The research in FY19 produced the following journal publications and patents: 1. Douglas C. Hofmann, Punnathat Bordeenithikasem, Zachary Dawson, Lee Hamill, Robert P. Dillon, Bryan McEnerney, Steven Nutt, Samuel C. Bradford. “Investigating bulk metallic glasses as ball-and-cone locators for spacecraft deployable structures.” Aerospace Science and Technology, 82-83, (2018 November). 2. Douglas C. Hofmann. “Dendrite-reinforced titanium-based metal matrix composites,” US20180339342A1 published November 29, 2018. The progress also included the procurement of two narrow size-ranges of spherical W powder, which was successfully integrated into the bulk metallic glass alloys to make composites with uniform microstructures. These composites were studied with both destructive and non-destructive techniques and appear appropriate for being used in the flight experiment.
Bibliography Type:	Description: (Last Updated: 01/01/2026)
Articles in Peer-reviewed Journals	Hofmann DC, Bordeenithikasem P, Dawson Z, Hamill L, Dillon RP, McEnerney B, Nutt S, Bradford SC. "Investigating bulk metallic glasses as ball-and-cone locators for spacecraft deployable structures." Aerospace Science and Technology. 2018 Nov;82-83:513-9. https://doi.org/10.1016/j.ast.2018.10.004 , Nov-2018
Patents	US 20180339342A1. Patent published November 29, 2018. http://www.freepatentsonline.com/y2018/0339342.html Nov-2018 Hofmann D. "Dendrite-reinforced titanium-based metal matrix composites."